

RESEARCH ARTICLE

Aerosol–Climate Interaction in the South Asian Region: A Focused Exploration of the Impacts of Ozone, Black Carbon, Optical Properties and Radiative Forcing in a Region of Significant Climate Change Dynamics

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Received: 2 October 2024 | **Revised:** 3 December 2024 | **Accepted:** 10 February 2025

Funding: This work was supported by Grants for Advanced Research in Education, Secondary and Higher Education Division, Ministry of Education (PS-2022-1971) and supported by the Department of Chemistry, University of Dhaka.

Keywords: aerosol optical depth | black carbon | heating rate | ozone | radiative forcing

ABSTRACT

In South Asia, our understanding of atmospheric aerosols and their optical properties is limited, posing a challenge to comprehending climate change dynamics. This study characterises aerosol optical properties, radiative properties, black carbon (BC) and ozone (O₃) at seven South Asian locations, including Nam Co (Tibetan Plateau, TP), Dhaka, Bhola (Bangladesh), and Hanimaadhoo, Kashidhoo, Male' and Gan (Maldives). The study utilises columnar aerosol data from the Aerosol Robotic Network (AERONET) and reanalysis data from Modern-Era Retrospective Analysis for Research and Applications (MERRA-2) from 2001 to 2020. Notably, during the winter, the highest Aerosol optical depth (AOD) levels were observed in Dhaka (1.0 ± 0.5) and Bhola (0.8 ± 0.4) among these seven locations. BC concentrations in Dhaka ranged from 2.1 to 2.8 μgm^{-3} , while Bhola recorded concentrations between 1.4 and 2.1 μgm^{-3} . O₃ levels across Maldives sites remained consistent, with values ranging between 314 and 345 dobson units (DU), surpassing those in Bangladesh and TP. The analysis shows a significant difference in the rate at which the atmosphere heats (HR) up due to aerosols. Higher heating rates were observed over Kashidhoo during the post-monsoon and winter seasons, while lower values were seen during the pre-monsoon and monsoon seasons, compared with Hanimaadhoo and Male'. It is important to note that Bangladesh had higher HR values than the Maldives. This study helps us better understand the impact of atmospheric aerosols on South Asia's climate and the different seasonal patterns.

1 | Introduction

The impact of aerosols on the climate system includes both direct effects (through scattering and absorption) and indirect effects (via changes in cloud properties), leading to significant uncertainties in radiative forcing assessments (Mehta 2015;

Rupakheti et al. 2020; Nair et al. 2023). One crucial optical parameter for quantifying atmospheric aerosol levels is the aerosol optical depth (AOD). Therefore, it is essential to conduct long-term global and regional investigations into AOD (Prasad et al. 2007; Mehta 2015; Nair et al. 2023). Aerosols exhibit temporal and spatial variability, which impacts the visual appearance

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of clouds. Their extensive presence in the atmosphere results in significant climatic effects due to interactions with clouds (Alam et al. 2010; Budhavant et al. 2016; Rupakheti et al. 2020). The optical and microphysical characteristics of columnar aerosols have been employed in various analyses to differentiate between different aerosol species, including absorbing aerosol forms. The impacts on weather, the radiative system and climate are primarily determined by the properties of columnar aerosols (Ramachandran and Rupakheti 2022). Ramanathan et al. (2005) demonstrated the radiative effects of aerosols in the Indian Ocean, resulting from their movement from contaminated continental source regions. These effects include modifications to the hydrological cycle and precipitation, highlighting the importance of understanding the consequences on climate (Persad 2023). However, further efforts are needed to quantify anthropogenic and natural aerosols' size and relative contributions (Budhavant et al. 2023). Furthermore, it is crucial to study the distinct processes that influence aerosol formation and the radiative characteristics of different aerosol components, for example, sulphate versus black carbon (BC) (Stier et al. 2007). Examining these factors will help minimise the uncertainties related to aerosols' radiative and climatic effects.

The Aerosol Robotic Network (AERONET) observation sites cover various land surface types, such as rural, urban, forest, desert, mountains and islands (Dubovik et al. 2000; Boiyo et al. 2019; Pokharel et al. 2019; Filonchik et al. 2021; Budhavant et al. 2018, 2024; Bilal et al. 2022). Pokharel et al. (2019) comprehensively evaluated aerosol optical characteristics over the Tibetan Plateau (TP) based on a decade of observations at two enduring AERONET sites, Nam Co and QOMS CAS. Zaman, Pavel, et al. (2021) investigated the yearly and seasonal patterns of AODs in major Bangladeshi cities. Similarly, Tariq et al. (2022) examined the long-term spatiotemporal trends of AOD using satellite remote sensing data in Pakistan. Mhawish et al. (2021) identified significant aerosol loading over the Indo-Gangetic Plain (IGP) and eastern India, noting that seasonal variations are linked to natural and anthropogenic sources. Ali et al. (2022) also reported a consistent increase in AOD from 2003 to 2020, attributed to anthropogenic emissions of BC, organic carbon and sulphate. Ramachandran and Rupakheti (2020) utilised in situ ground-based observations over Hanimaadhoo and the nearby resource areas in 2014, conducting the first perennial comprehensive inquiry into crucial climate-related aerosol factors. This included content and structure, absorption due to dust, BC, brown carbon (BrC), radiative forcing, efficiency and resultant heating rates (HR). BC significantly alters atmospheric stability by absorbing and scattering solar radiation (Pavel et al. 2023; Nair et al. 2023, 2024). It affects large-scale circulations, cloud albedo and the formation of ice and cloud nuclei (Johnson et al. 2019; Budhavant et al. 2023). BC also influences local and global radiation balance and climate (Ramanathan and Carmichael 2008; Bond et al. 2013; Nair et al. 2023). In regions marked by high levels of BC pollution, conducting a comprehensive analysis of BC sources and implementing effective mitigation measures is crucial. This is imperative due to BC's substantial detrimental environmental and human health impacts.

Asia is the world's largest continent, home to 60% of the global population and occupies 30% of the world's land area. Despite

the growing body of evidence from numerous global and regional studies (Joy et al. 2024; Zaman, Pavel, et al. 2021; Zaman et al. 2022; Budhavant et al. 2024), a significant gap persists in our comprehensive understanding of aerosol characteristics, especially their optical and radiative properties across the diverse landscapes of South Asia. This region, characterised by a wide range of climatic and geographical conditions, has not been thoroughly examined regarding the complexities of aerosol behaviour. A particularly critical knowledge gap lies in the scarcity of detailed studies focusing on aerosols' seasonal variability and how these fluctuations impact local and regional climates. Such a lack of information hampers our ability to grasp the broader implications of climate dynamics and poses challenges to effective air quality management in the region.

Our research addresses the vital need for systematic and comparable long-term studies on aerosol optical and radiative properties, BC levels and total ozone column measurements across seven strategically located locations in South Asia. The innovative nature of our work lies in its comparative approach, highlighting the effects of pollution from high-emission source areas, like Dhaka and Bhola, on remote receptor sites such as Hanimaadhoo, Kashidhoo, Male', Gan and Nam Co (Figure 1A). We have set forth several ambitious objectives: to analyse aerosol optical depth (AOD) across all seven locations using reanalysis data; to explore seasonal variations in the Angström exponent (AE) and fine mode fraction (FMF) through ground-based observations; to examine concentrations of BC and ozone; and to assess aerosol radiative forcing in selected sites in the Maldives and Bangladesh. By blending ground-based observations with reanalysis data, our research unveils valuable insights into aerosol radiative forcing. This comprehensive approach deepens our understanding of aerosols' intricate effects on climate systems. The findings broaden knowledge in this field and have profound implications for enhancing regional climate models and shaping effective environmental policies.

2 | Methodology

2.1 | Measurement Sites

This analysis was conducted at four AERONET sites in the Maldives, specifically Kashidhoo, MCO-Hanimaadhoo, Male' and Gan. The Maldives is an archipelago of coral islands stretching 800 km from north to south in the Indian Ocean (Budhavant et al. 2023). The capital and largest city of the Maldives, Male', has a population of 252,768 and a terrestrial area of 8.30 km². Kashidhoo is surrounded by deep waters with strong surf, featuring a thin, bordering reef on its northwest side and an oval-shaped lagoon. Gan, the southernmost island of the Maldives and the southernmost island of Addu Atoll, spans 2.2561 km² and is the second-largest island in the atoll (Budhavant et al. 2024). Based on South Asian meteorological conditions, the seasons in the Maldives can be categorised into pre-monsoon, monsoon, post-monsoon and winter.

Bangladesh hosts two AERONET sites in Dhaka and BCOB (Bhola Climate Observatory-Bangladesh). BCOB is situated in the Bhola district, and one AERONET site of TP is Nam Co (Figure 1A). Bangladesh, a Southeast Asian republic, is

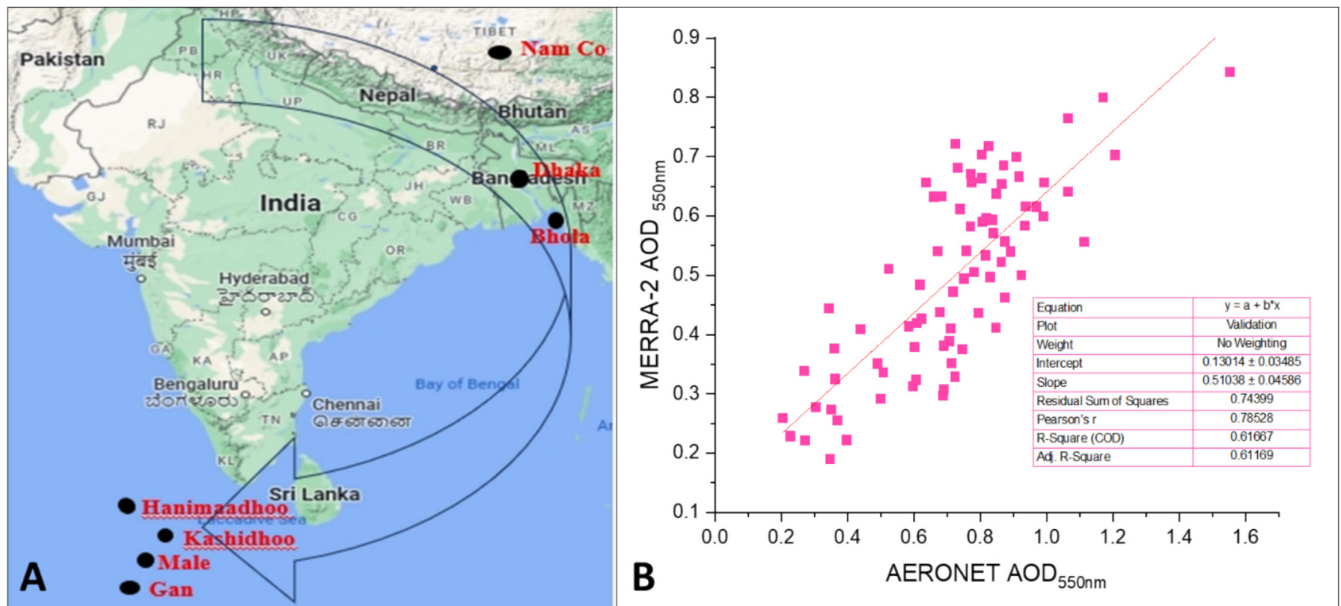


FIGURE 1 | (A) Map of study locations including Nam Co on the Tibetan Plateau, Dhaka and Bhola in Bangladesh and Hanimaadhoo, Kashidhoo, Male' and Gan in the Maldives (Source: Google Maps). (B) Correlation between ground-based AOD (AERONET) and reanalysis model-based AOD (MERRA-2). [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

surrounded by Myanmar to the southeast, India to the south-east and the Bay of Bengal (BoB) to the southwest. It is one of the most densely populated nations, covering an area of about 57,320 miles² with a population of ~160 million (Zaman et al. 2022). Bangladesh experiences four distinct seasons based on the average temperature and precipitation over 3 months: pre-monsoon (March–May), monsoon (June–August), post-monsoon (September–November) and winter (December–February) (Salam et al. 2012; Pavel et al. 2021; Zaman, Yesmin, et al. 2021). Dhaka serves as the capital and commercial centre of Bangladesh. The remote island district of Bhola in extreme southern Bangladesh is near the Bay of Bengal (Zaman et al. 2022). Nam Co (4746 masl, 30.77°N, 90.96°E) is a high-altitude scientific research observatory focusing on the interactions between various spheres, such as the atmosphere, hydrosphere, cryosphere and environmental changes. It is situated between Nam Co Lake and the Nyainqentanglha Mountains in the Tibetan Plateau. The nearby alpine grassland features predominantly flat topography and is characterised by lakes and mountain glaciers. This area has a small population and no industries due to its high elevation and harsh climate (Pokharel et al. 2019).

2.2 | Data Sources

2.2.1 | AERONET Data

The AERONET data is widely trusted and used for ground-based aerosol monitoring worldwide. It utilises a programmed sky/sun scanning radiometer (CIMEL 318) to measure solar radiation every 5 min, providing a complete field of view at various wavelength channels ranging from 340 to 1020nm, including a 940nm water vapour band (Holben et al. 1998; Pokharel et al. 2019). The range of the estimated AOD's uncertainty is around 0.01–0.02. For sky irradiance reflection, which depends on the types and quantities of aerosols, it is <5% (Dubovik

et al. 2000; Mehta 2015; Zaman, Pavel, et al. 2021). Level two and version three cloud-separated and quality-secured day-to-day data items were employed in this analysis. Aerosol optical depth (AOD_{500nm}) and radiative properties like aerosol radiative forcing (ARF), aerosol radiative forcing efficiency (ARFE) below the atmosphere (BOA), in the atmosphere (ATM) and at the top of the atmosphere (TOA) were used. The aerosol heating or radiative forcing (Wm⁻²) in the atmosphere (ARF_{ATM}) was used to evaluate the periodic mean atmospheric solar rate of heating (HR) for each season. The fine mode fraction (FMF) is computed from measurements of sky radiance since the segment of fine mode AOD to entire AOD (because of all elements in the range of 0.01–10μm radius) (Ramachandran and Rupakheti 2020). Spectral AOD measurements in the 0.44–0.87μm wavelength region are used to calculate the Angström exponent. The uncertainty in AODs calculated from measurements of direct sun radiation is less than ±0.01 for wavelengths higher than 0.44μm and a lesser extent than ±0.02 for fleeting wavelengths (Holben et al. 2001). The inaccuracy in the FMF that AERONET deduced is 10% (O'Neill et al. 2003). Many data points were lost due to equipment errors and severe rain during the monsoon seasons (Zaman et al. 2022).

2.2.2 | MERRA-2 Reanalysis Data

NASA's Worldwide Modelling and Integration Headquarters undertook the Modern-Era Retrospective Analysis for Research and Applications (MERRA) with two main goals in mind: to better represent the hydrologic cycle in earlier generations of reanalysis and to contextualise remarks from NASA's Earth Observing System satellites (Rienecker et al. 2011). The most recent version of the MERRA reanalysis dataset is MERRA-2 (Rienecker et al. 2011; Ali et al. 2022). Multiple satellite-built AOD outputs from the AVHRR, MISR and MODIS devices are used by MERRA-2. Additionally, AERONET data from the

ground are incorporated (Ali et al. 2022). With a spatial resolution of 0.5×0.625 , MERRA-2 reanalysis AOD, Ozone and BC records were retrieved from (<https://giovanni.gsfc.nasa.gov/giovanni/>, accessed September 10, 2022). One of the major limitations of MERRA-2 data is that it does not consider the aerosol nitrate AOD (Randles et al. 2017; Song et al. 2018; Rawat et al. 2019).

2.3 | Research Methodology

The link between ground-constructed data of AERONET AOD₅₀₀ and reanalysis-based MERRA-2 data of AOD₅₅₀ was assessed to gain a comprehensive picture of the aerosol optical depths over the seven South Asian locations. It is pivotal to compare each sensor to a standard wavelength when comparing AERONET and MERRA-2 results; hence, the AERONET AOD at 500nm has been changed to 550nm (1999; Levy et al. 2005; Zaman, Pavel, et al. 2021). The approach was used in earlier analyses (Alam et al. 2011; Rupakheti et al. 2018; Zaman et al. 2022; Bibi et al. 2015) applying the following formula:

$$\text{AOD}_{550\text{nm}} = \text{AOD}_{500\text{nm}} (550/500)^{-\alpha} \quad (1)$$

where α is the Angström exponent (440–870 nm).

The Pearson correlation (r) value of 0.78 and regression value of 0.62 (Figure 1B) between MERRA-2 AOD (550nm) and AERONET AOD (550nm) indicate a good correlation.

This study focuses on the influence of aerosol loading on the energy balance (Rupakheti et al. 2021). The radiative forcing data from the AERONET website analysed diverse locations such as the Maldives and Bangladesh. The study encompassed various scenarios, including urban-industrial, mineral dust, biomass burning, marine aerosols, background continental and unrestricted troposphere. The solar fluxes measured by AERONET were correlated with ground-based measurements, showing a relationship of over 99% (García et al. 2012; Ramachandran and Rupakheti 2020). Atmospheric radiative forcing data was collected both at the top of the atmosphere (ARF_{TOA}) and below the atmosphere (ARF_{BOA}), distinguishing between the two

as atmospheric radiative forcing (ARF_{ATM}) (Ramachandran and Rupakheti 2020). Additionally, the following equation (Equation 2) was used (Rupakheti et al. 2019) to estimate the HR induced by aerosols to understand the impact of dust episodes in these locations.

$$\frac{\delta T}{\delta t} = \frac{g}{C_p} \times \frac{\Delta F(\text{ATM})}{\Delta P} \times 24 \left(\frac{\text{Hour}}{\text{Day}} \right) \times 3600 \left(\frac{\text{Sec}}{\text{Hour}} \right) \quad (2)$$

where $\frac{\delta T}{\delta t}$ is the HR of the atmosphere induced by aerosol in Kelvin day⁻¹, g (9.8 ms^{-2}) is the gravitational acceleration, C_p stands for the ambient air's specific heat capacity at constant pressure ($1006 \text{ J kg}^{-1} \text{ K}^{-1}$), $\Delta F(\text{ATM})$ is the ARF_{ATM} (in W m^{-2}). ΔP (hPa) is the atmospheric pressure difference between the surface and a height of 3 km over the measurement sites. The typical atmospheric aerosols are found inside 3 km from the surface; hence, 3 km is used as the distance threshold.

3 | Results and Discussion

3.1 | Long-Term Analysis of AOD Over South Asia by Using Reanalysis Data

Aerosol optical depth (AOD) measures how much light is absorbed or scattered by aerosol particles in the atmosphere. It is widely used in atmospheric research and environmental monitoring to understand air quality, climate change and the impact of aerosols on visibility and human health. The examination of MERRA-2 reanalysis data reveals that Bangladesh exhibits a higher concentration of AOD than other research locations in South Asia.

The AOD levels range from 0.38 to 0.50 in Dhaka and from 0.33 to 0.48 in Bhola, showing significant annual variations, as depicted in Figure 2A. Our research reveals that in 2012, the highest AOD value in Dhaka was recorded at 0.50 ± 0.16 , while in Bhola, it reached 0.48 ± 0.14 in 2018. Conversely, the Nam Co region exhibits the lowest AOD trend, with values ranging from 0.05 to 0.07, suggesting minimal AOD fluctuation. Additionally, in the Maldives, including Hanimaadhoo, Kashidhoo, Male and

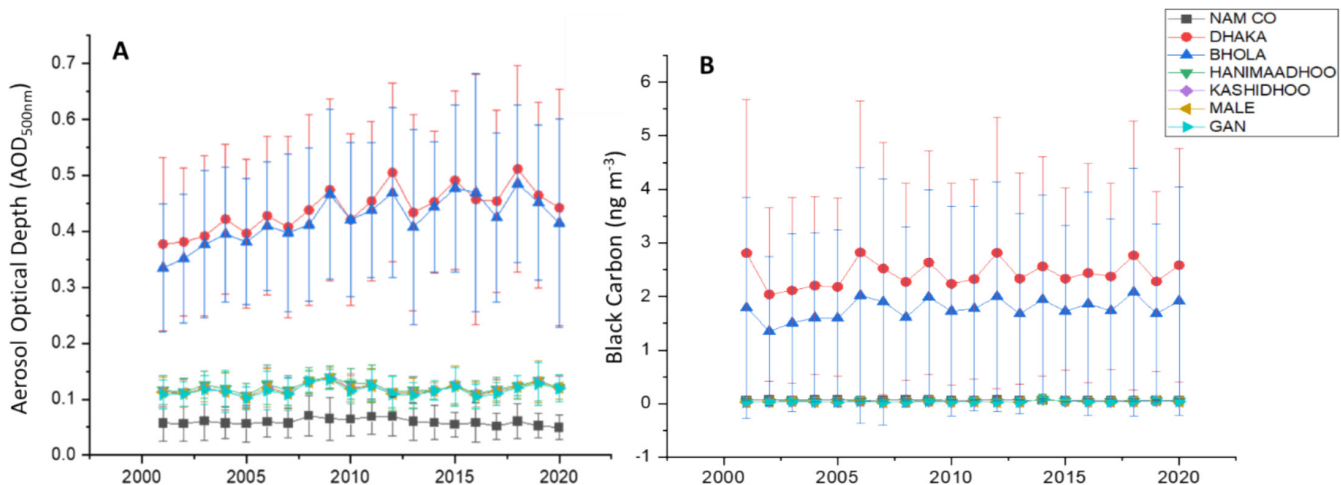


FIGURE 2 | (A) Long-term trend analysis of aerosol optical depth (AOD) and (B) variation of black carbon (BC) concentration at seven different locations in South Asia using reanalysis data from MERRA-2 (2001–2020). [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

Gan, AOD trends show remarkable similarities, ranging from 0.10 to 0.14. The elevated AOD levels observed in Bangladesh suggest that the region experiences a substantial presence of anthropogenic aerosols, which can impact air quality and climate. The similarity in AOD levels between Dhaka's vibrant city and Bhola's more tranquil coastal region can be attributed to several intriguing factors. The transboundary aerosol transport from the IGP is a significant contributor. During the winter, north-westerly winds carry a substantial load of pollutants into Bhola, a noteworthy occurrence, given its relatively isolated location. Moreover, the high humidity typical of coastal environments like Bhola is vital in forming secondary aerosols. This process occurs when moisture in the air interacts with various pollutants, resulting in elevated AOD levels. Additionally, seasonal meteorological conditions, particularly during winter, exacerbate this phenomenon. Temperature inversions—where warmer air traps cooler air near the surface—combined with low wind speeds create an environment where aerosols can quickly accumulate. Consequently, both Dhaka and Bhola experience heightened levels of aerosol pollution.

3.2 | Angström Exponent (AE) and Fine Mode Fraction (FMF)

The AE indicates the spectral dependency of AOD concerning the predominant dimensions of the aerosol size distribution (Giles et al. 2012; Ramachandran and Rupakheti 2022). A higher AE signifies the prevalence of fine-mode particles in the aerosol size distribution, whereas a lower AE suggests the dominance of larger aerosols (Ramachandran and Rupakheti 2022). Notably, the AE values over Nam Co remain consistently below 1.0 throughout the year, as depicted in Figure 3A. This implies that coarse particles, such as dust and sea salt, dominate the aerosol dispersion in this region. The immense AE value (0.82) was observed during winter over Nam Co. Conversely, in Dhaka, AE values consistently exceed one during all seasons, decreasing to 1.10 during the monsoon. This suggests the predominant involvement of fine-mode particles in AOD in this location for most of the year.

In Bhola, a coastal area of Bangladesh, Hanimaadhoo and Kashidhoo, AE values surpass 1 in all seasons except the monsoon. During the monsoon, the AE values are 0.93, 0.44 and 0.54 over Bhola, Hanimaadhoo and Kashidhoo, respectively, with Bhola exhibiting nearly twice the AE compared with Hanimaadhoo. The elevated AE values in these locations during winter, pre-monsoon and post-monsoon can be attributed to the northeastward transport of fine-mode aerosols released by anthropogenic events, such as biomass burning and fossil fuel combustion, especially over the contaminated IGP in South Asia (Ramachandran and Rupakheti 2020). Over Male, AE consistently exceeds one during pre-monsoon and winter, while over Gan, AE remains below one throughout all seasons.

The FMF provides quantitative insights into the characteristics of aerosol size distribution, ranging from zero (indicating a single coarse mode aerosol) to one (indicating a single fine mode aerosol) (Kedia et al. 2014). Across Nam Co, Dhaka, Bhola, Hanimaadhoo, Kashidhoo, Male and Gan, the FMF values exhibit seasonal variations within the ranges of (0.4–0.6), (0.7–0.9), (0.6–0.9), (0.3–0.9), (0.4–0.7), (0.3–0.8) and (0.3–0.5), respectively. Notably, FMF values are lower during the monsoon and higher in winter. This decrease in FMF during the monsoon is attributed to an upsurge in dust particle concentration (Eck et al. 2010; Giles et al. 2011; Kedia et al. 2014). The elevated FMF values during winter can be attributed to multiple factors. First, high anthropogenic emissions impact the entire IGP during winter (Di Girolamo et al. 2004; Singh et al. 2004; Prasad et al. 2007; Eck et al. 2010; Giles et al. 2011) trapping aerosols transmitted from west to east due to northwesterly winds and a lower boundary layer altitude (Dey and Di Girolamo 2010). Additionally, extensive burning of agricultural residue, particularly in Dhaka and Bhola, contributes to the heightened FMF values (Badarinath et al. 2009).

3.3 | Trends in BC and Ozone

Between 2001 and 2020, our exploration revealed a consistent variation in BC levels in Dhaka, ranging from 2.0 to 2.90 $\mu\text{g m}^{-3}$

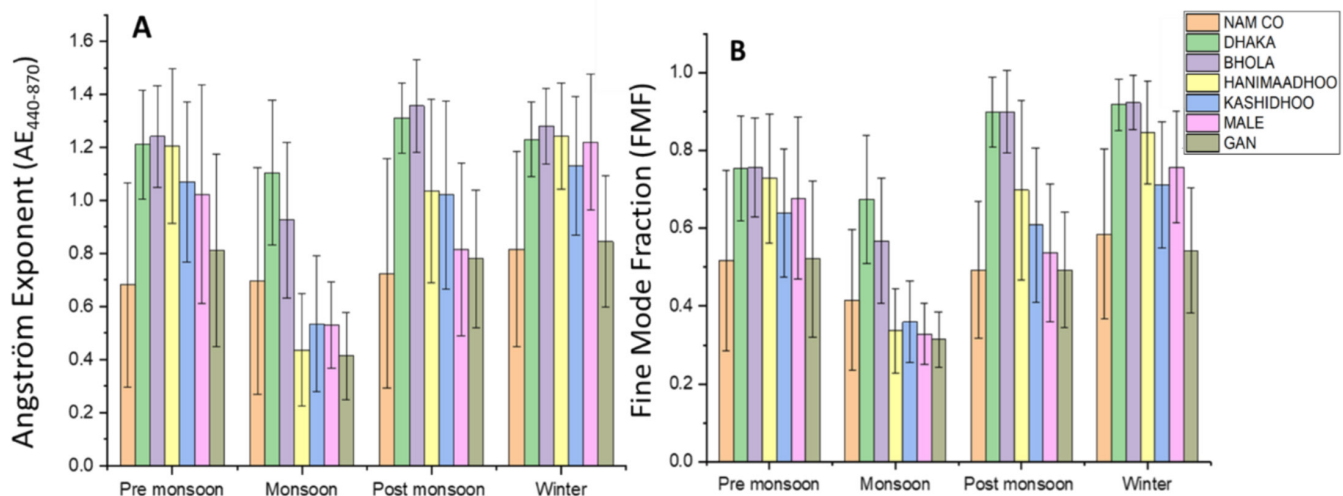


FIGURE 3 | (A) Seasonal variation of the Angström exponent and (B) Seasonal mean of fine mode fraction (FMF) at seven locations in South Asia. Vertical bars represent the standard deviation. [Colour figure can be viewed at wileyonlinelibrary.com]

throughout the year, surpassing concentrations observed in other research locations. Except for 2006, BC levels in Bhola remained below $2 \mu\text{g m}^{-3}$ but consistently exceeded $1 \mu\text{g m}^{-3}$. Notably, BC values in Dhaka consistently exceeded $2 \mu\text{g m}^{-3}$. In Nam Co, readings fluctuated between 0.06 and $0.07 \mu\text{g m}^{-3}$, representing a minimum of 167 times lower values than those observed in Dhaka. The study locations in the Maldives consistently exhibited a regular pattern in BC values throughout the observation period, averaging $0.04 \mu\text{g m}^{-3}$ (Figure 2B). Given the higher trend in BC levels in Dhaka and Bhola compared with the other four locations, these areas emerge as the primary sources of BC emissions. In contrast, the other four locations in the Maldives function as receptor sites for BC. The typical atmospheric lifespan of BC is 7 days (Bond et al. 2013; Qin et al. 2019). Consequently, the long-distance transport of BC in air masses from nearby areas may contribute to BC concentrations at receptor locations (Qin et al. 2019). The incomplete incineration of fossil fuels and biomass is a significant source of BC pollutants. The leading causes of BC pollution include household emissions, transportation, industrial activities, power plants and open burning of biomass. BC pollution is most pronounced in Dhaka due to substantial transportation, industrial activities and household practices. Conversely, in the case of Bhola, burning biomass is the main factor driving higher BC levels.

Figure 4A depicts the ozone levels observed across different locations throughout the study period. It is crucial to highlight that the presented ozone measurements represent the total column ozone, encompassing ozone distributed across various atmospheric layers. The interpretation of these observations should acknowledge the intricate vertical ozone distribution in the atmosphere, with a substantial portion in the stratosphere. Throughout the observation period, ozone levels in the Maldives' Hanimaadhoo, Kashidhoo, Male and Gan consistently followed a similar trajectory, fluctuating between 314 and 345 dobson units (DU). In contrast, Dhaka and Bhola exhibited ozone levels ranging from 262 to 275 DU, with readings at the Nam Co site aligning closely with those in Dhaka and Bhola.

The variations in total column ozone levels observed across different locations can be influenced by myriad factors,

encompassing regional emissions of ozone precursors, long-range transport and atmospheric circulation patterns. While the data may suggest higher ozone levels in the Maldives sites compared with Dhaka, Bhola and Nam Co, it is crucial to avoid simplistic interpretations that imply a direct link between ground-level processes and total column ozone. Understanding ozone distribution and its variation across regions is paramount for comprehensively assessing atmospheric dynamics and air quality. Further investigations are required to investigate the complex interplay of factors affecting ozone levels in these diverse locations.

3.4 | Characterisation of Aerosol Radiative Properties

Figure S1 elucidates the aerosol radiative forcing (ARF) for various aerosol events at three stages: at the top of the atmosphere (ARF_{TOA}), at the bottom of the atmosphere (ARF_{BOA}) and within the atmosphere (ARF_{ATM}). Negative values at TOA indicate air cooling caused by light dispersed back into space, while negative values at BOA imply cooling resulting from a reduction in solar energy reaching the Earth's surface or the darkening of the surface itself (Kaskaoutis et al. 2007). The average ARF_{TOA} , ARF_{ATM} and ARF_{BOA} values during different seasons in Kashidhoo were as follows: Pre-monsoon: -18.7 , 27.1 , -45.7 W m^{-2} ; Monsoon: -11.2 , 10.8 , -22.0 W m^{-2} ; Post-monsoon: -8.5 , 13.7 , -22.2 W m^{-2} ; Winter: -10.8 , 24.7 , -35.5 W m^{-2} . In Male, the highest ARF_{TOA} value occurred in the winter season (-20.3 W m^{-2}), with the highest ARF_{ATM} and ARF_{BOA} values (19.2 W m^{-2} , -39.5 W m^{-2}) also observed in winter. Hanimaadhoo exhibited the highest ARF values in ATM and BOA during the winter season (23.0 , -47.4 W m^{-2}), while the highest TOA value was found in the pre-monsoon season (-26.0 W m^{-2}). Ramachandran and Rupakheti (2020) also calculated ARF values for Hanimaadhoo, considering only 1 year (2014). Their study revealed the highest ARF_{BOA} and ARF_{TOA} in the post-monsoon season, while the highest ARF_{ATM} was observed in winter. For Dhaka and Bhola, ARF values in BOA, TOA and ATM were higher in winter (Dhaka: -132.9 , -43.4 , 89.5) and (Bhola: -108.1 , -30.8 , 77.2).

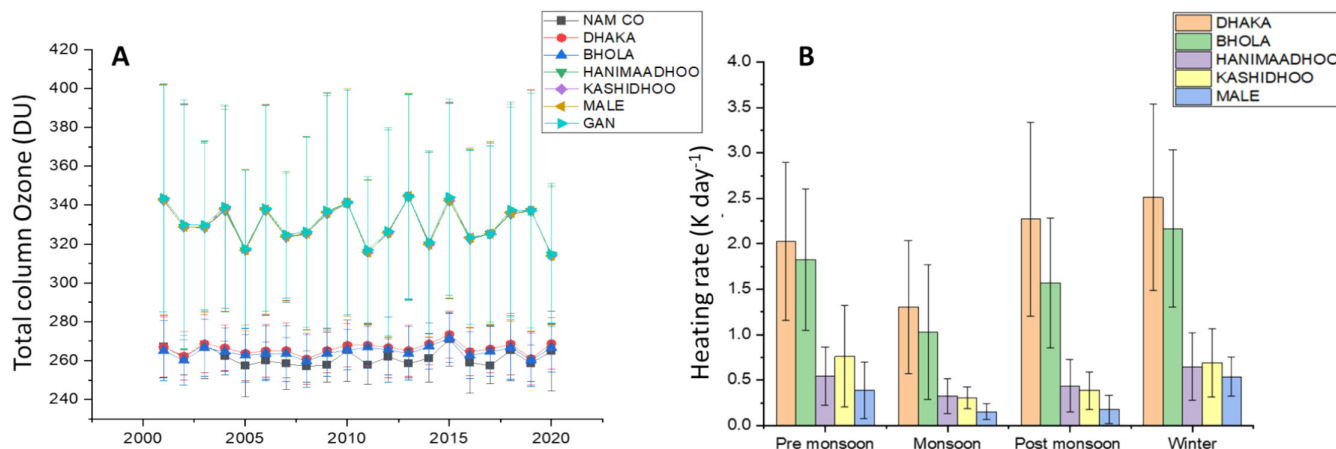


FIGURE 4 | (A) Atmospheric ozone levels in South Asia from 2001 to 2020 at seven locations. (B) Evaluation of aerosol heating rate at Dhaka, Bhola, Hanimaadhoo, Kashidhoo and Male'. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

Whereas their radiative forcing in the troposphere and at the upper level of the atmosphere varies, both scattering and absorbing aerosol types result in cooling, or negative ARF, of the Earth's surface. The ARF_{TOA} and ARF_{BOA} in the shortwave region are similar for scattering aerosols (greater SSA, e.g., sulphate-dominated). In contrast, the ARF_{BOA} for absorbing aerosols (minor SSA, e.g., BC dominated) is roughly 2–3 times greater than ARF_{TOA} in the shortwave section, which results in more substantial atmospheric warming. ARF_{BOA} and AODs are linearly connected for aerosols of the same kind (i.e., SSA). The ARF_{TOA} , on the other hand, is dependent on the SSA, and on ocean surfaces (little surface albedo), it gets worse as the SSA rises (Ramachandran and Rupakheti 2020). The average value of ARF at the ATM was higher in Dhaka compared with the other four locations and lower in Male, underscoring the importance of understanding aerosol behaviour in different regions. This knowledge is crucial for accurately assessing the impact of aerosol events on atmospheric heating and developing effective mitigation strategies. Our findings will be valuable to scientists, researchers and academics in atmospheric science and environmental studies.

The aerosol radiative forcing efficiency (ARFE), defined as the ARF per unit AOD ($W m^{-2} AOD^{-1}$), provides insights into the impact of aerosol absorption and scattering properties (García et al. 2008; Rupakheti et al. 2021). The calculated average values of ARFE at the atmosphere (ATM), top of the atmosphere (TOA) and bottom of the atmosphere (BOA) in Kashidhoo (Figure S2) were 27.1, -18.7 and $-184 W m^{-2} AOD^{-1}$, respectively, during the pre-monsoon season. However, the maximum ARFE at BOA was observed in winter, reaching $-230 W m^{-2} AOD^{-1}$. In Male, the capital of the Maldives, ARFE at ATM was highest in winter ($83.0 W m^{-2} AOD^{-1}$) and lowest in the monsoon ($45.5 W m^{-2} AOD^{-1}$). Other peak values of ARFE at BOA and TOA were -171 (winter) and -106 (monsoon), respectively. In the location of Hanimaadhoo, the highest average values of ARFE at ATM, TOA and BOA were 77 (winter), -86 (monsoon) and -156 (winter), respectively. The ARFE values in Dhaka and Bhola at BOA and TOA were higher for both sites in the post-monsoon and monsoon seasons, respectively. Additionally, the ARFE at ATM peaked in the post-monsoon season (Dhaka: 126 and Bhola: 138) compared with other seasons. Notably, the ARFEs during the monsoon were lower over Hanimaadhoo when compared with other seasons (Ramachandran and Rupakheti 2020).

The HR across all locations exhibited higher values during the post-monsoon and winter seasons but lower values in the pre-monsoon and monsoon seasons (Figure 4B). Dhaka and Bhola consistently demonstrated higher HR than the Maldives sites, with Kashidhoo having the highest average HR among the Maldives sites. The average HR values were Dhaka ($2.0 \pm 0.9 K day^{-1}$), Bhola ($1.7 \pm 0.8 K day^{-1}$), Hanimaadhoo ($0.5 \pm 0.3 K day^{-1}$), Kashidhoo ($0.5 \pm 0.3 K day^{-1}$) and Male ($0.3 \pm 0.2 K day^{-1}$) (Table S1). Over Hanimaadhoo and Male, HR caused by aerosols was more pronounced during the winter, with values of 0.7 and $0.5 K day^{-1}$, respectively. During other seasons, HR decreased, reaching the lowest values during the monsoon over both Hanimaadhoo and Male. In Kashidhoo, higher HR was observed during the pre-monsoon season compared with winter, despite the relatively small HR values fluctuating between

the two seasons ($0.8 K day^{-1}$ in pre-monsoon and $0.7 K day^{-1}$ in winter). The lowest HR value was recorded during the monsoon season ($0.3 K day^{-1}$). Similarly, HR was lower in the monsoon for both Dhaka and Bhola.

Several studies were carried out in different locations, and the observed HR values were comparatively higher: 0.8 – $2.7 K day^{-1}$ over Kanpur in India (Kumar et al. 2015; Tiwari et al. 2019), 1.3 – 2.2 and 0.7 – $1.4 K day^{-1}$ over Lahore and Karachi in Pakistan (Iftikhar et al. 2018), 0.89 – $2.16 K day^{-1}$ over Zanzan site in Iran (Gharibzadeh et al. 2017), 2.4 – $3.8 K day^{-1}$ over the capital of China (Beijing) (Yu et al. 2016), 5.0 – $7.0 K day^{-1}$ over Lumbini in Nepal (Rupakheti et al. 2020) and 0.8 – $2.7 K day^{-1}$ over Dushanbe in Central Asia (Rupakheti et al. 2021). Taken together, this comprehensive study provides valuable insights into how atmospheric aerosols affect the climate of South Asia across different seasons. The research sheds light on the intricate seasonal patterns influenced by these aerosols, contributing to a more thorough understanding of their impact on the region's climate.

4 | Conclusion

A comprehensive study conducted between 2001 and 2020 analysed aerosol optical properties in seven South Asian locations. Throughout South Asia, there are notable variations in the atmosphere's composition. In particular, Dhaka and Bhola stand out with a higher trend in AOD than other locations under study. Additionally, the Maldives consistently experience higher ozone levels than Bangladesh and the Tibetan Plateau. Furthermore, aerosols in Bangladesh contribute to a higher atmospheric heating rate than the Maldives.

The findings revealed that Dhaka and Bhola had significantly higher levels of aerosols, indicating more significant pollution than other sampling sites. The incomplete incineration of fossil fuels and biomass is a primary source of BC pollutants in Bangladesh. Due to significant transportation, industrial activities and household practices, BC pollution is particularly severe in Dhaka. Conversely, in the case of Bhola, the main factor driving higher BC levels is biomass burning. Conversely, the sites in the Maldives exhibited lower levels of aerosols. The study also observed stable levels of BC in the Maldives, while Dhaka and Bhola showed higher levels, suggesting substantial emissions. It is essential to note the study's emphasis on considering the limitations when interpreting the data. Taken together, this study helps us better understand the impact of atmospheric aerosols on South Asia's climate and the different seasonal patterns.

Author Contributions

Kazi Rakib Uddin: conceptualization, methodology, data curation, formal analysis, writing – review and editing, writing – original draft, investigation, software. **Shahid Uz Zaman:** conceptualization, methodology, data curation, supervision, formal analysis, investigation, writing – original draft, writing – review and editing, software. **Krishnakant B. Budhavant:** writing – review and editing, formal analysis, writing – original draft, investigation. **Abdus Salam:** conceptualization, methodology, investigation, writing – review and editing, writing – original draft, funding acquisition, project administration, supervision.

Acknowledgements

The authors acknowledge the cordial support of NASA, USA, for providing different CIMEL sun photometers to operate different AERONET stations in study locations. We also thank Giovanni (nasa.gov) for providing reanalysis (MERRA-2) data.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that supports the findings of this study are available in the Supporting Information material of this article.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.